THE HEFAISTOS: Academic & Policy Assessment Platform

> **SPECIFICATION:** Bridging Macroeconomic Computable General Equilibrium (CGE) Modeling, Geophysical Climate Science, and Multi-Criteria Global Equity

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Research Monograph Series: Sampo Deep-Tech Technical Report Series (`HF-TR-2026-V2.0`)

Compliance & Methodological Standards: Finnish Meteorological Institute (Partanen et al. 2026, ERL) · IPCC AR6 WG1/WG3 · IEA Energy Statistics · UNFCCC Machine-Validated NDCs · ASME V&V 40 Computational Standard Aligned

| KEY PERFORMANCE METRICS

153 Countries	841 Ensembles	5 Dimensions	100% Deterministic
Global Earth System & Macro CGE Coverage	FaIR v2.2.4 Climate Distributions ($k_{\text{norm}} = 1.04$)	Multi-Criteria Climate Justice & Burden-Sharing Matrix	Zero-Hallucination Numerical Compute Core & Audit Trail

NOTICE OF INTELLECTUAL PROPERTY & ARCHITECTURAL SCOPE

This specification constitutes formal academic and policy research documentation released under the Sampo Deep-Tech Technical Report Series (`HF-TR-2026-V2.0`). The scope of this technical disclosure is strictly constrained to mathematical carbon budget adjustment formulations, general equilibrium production elasticities, geophysical warming response functions, and multi-criteria equity matrix algorithms. In strict accordance with institutional governance, this document intentionally omits proprietary API authentication tokens, production database connection strings, uncompiled numerical optimization solvers for large-scale non-linear Pyomo routines, and internal cryptographic private keys. All equations, parameter sets, and architectural schemas disclosed herein are provided exclusively for scientific peer-review, policy assessment, and multilateral regulatory audit under UNFCCC, IPCC, and EU climate frameworks.

| EXECUTIVE SUMMARY & SYSTEM ARCHITECTURE

Global climate governance stands at a critical juncture. Decarbonization policies have historically suffered from deep architectural silos: macroeconomic models evaluate fiscal policy without physical climate feedback; climate pledges (NDCs) celebrate paper targets while ignoring accounting discrepancies; and international negotiations stall over rigid, one-dimensional equity models.

The Hefaistos is a next-generation **Integrated Academic & Policy Assessment Platform**. It connects macroeconomic Computable General Equilibrium (CGE) modeling, empirical energy engineering, geophysical climate carbon budgets (FaIR v2.2.4 ensemble modeling), and multi-criteria climate justice into a unified, deterministic, and fully reproducible analytical engine.

Covering **153 countries globally**, The Hefaistos provides ministers, climate negotiators, and researchers with publication-ready, auditable evidence packages that quantify not only economic costs and emissions trajectories, but the **exact physical warming response (ΔT in millikelvins)** and **remaining national carbon budgets** across five distinct equity paradigms.



1. THE PROBLEM LANDSCAPE: WHY TRADITIONAL MODELS FAIL

Current climate policy decision-making is severely constrained by three foundational methodological flaws:

1.1 The "Paper Ambition" vs. Physical Reality Disconnect

National greenhouse gas inventories (NGHGI) reported to the UNFCCC use accounting rules that differ systematically from global earth system models:

Unmanaged Forest Sinks: NGHGI conventions credit countries for natural, non-anthropogenic sink enhancements driven by environmental changes (e.g., CO_2 fertilization, climate warming), whereas global carbon budgets define net zero purely on direct human-induced flows (Grassi et al., 2021).

Short-Lived Climate Pollutants (SLCPs): Standard GWP100 metrics misrepresent the climate impact of methane (CH_4). Rapid methane reductions produce near-term cooling, whereas steady emissions sustain warming without cumulative atmospheric accumulation.

1.2 Macroeconomic Blindness to Physical Constraints

Traditional macroeconomic models (CGE and DSGE) evaluate carbon taxes and investment subsidies based purely on GDP growth, employment shifts, and percentage emissions cuts. They operate in a vacuum, unable to answer the ultimate scientific question: **"Does this national policy pathway breach our geophysical 1.5 °C carbon budget, and by how much?"**

1.3 Equity Deadlocks in Climate Negotiations

International burden sharing has been crippled by binary disputes: developing nations demand historical per-capita reparations, while developed nations advocate for capacity-based (ability-to-pay) scaling. Crucial structural factors—such as **basic-needs heating in cold climates (HDD)** and **carbon leakage via embodied carbon in global trade**—have been excluded from quantitative tools.

2. CORE ARCHITECTURAL PILLARS & ENGINES

The Hefaistos resolves these challenges through five tightly coupled, deterministic computational engines:

THE HEFAISTOS ENGINE SUITE	
1. CGE Engine	Calibrated General Equilibrium macro solving, CES production, carbon tax response (€/tCO2), and sectoral employment/output.
2. FMI Climate Engine	Implements Partanen et al. (2026): A_LULUCF, A_nonCO2GHG, CO2-we, FaIR v2.2.4 sub-additivity factor k=1.04, and TCRE transient warming.
3. Equity Engine	5-dimensional burden sharing: Ability-to-Pay, Equal Per-Capita, Historical (1990+), HDD Heating basic-needs, and Trade Consumption.
4. Engineering Engine	VRE capacity (Solar/Wind GW), Storage proxies, Electrification rates, Grid upgrade indices, and empirical coal phase-out trend estimation.
5. NDC Gap Engine	Machine-validated NDC extraction vs Paris 1.5°C / 2.0°C trajectories.

3. MATHEMATICAL & METHODOLOGICAL FORMALISM

3.1 Physical Carbon Budget Adjustment Framework

Following Partanen et al. (2026), the global remaining carbon budget ($B_{\text{gl, GHG}} = 936.0$ Gt CO_2-eq) is adjusted to align national inventory accounting (NGHGI) with global climate dynamics:

$$B_{\text{nat, GHG, NGHGI}} = f \cdot B_{\text{gl, GHG}} + A_{\text{LULUCF, nat}} + A_{\text{nonCO2GHG, nat}}$$

Where:

LULUCF Indirect Sink Adjustment (A_{LULUCF}):

$$A_{\text{LULUCF, nat}} = \int_{2020}^{2050} \left(\text{Sink}_{\text{NGHGI, nat}}(t) - \text{Sink}_{\text{BM, nat}}(t)\right) dt$$

Corrects for the discrepancy between bookkeeping models (BM) and national managed land reporting.

Non-CO_2 Warming-Equivalent Adjustment ($A_{\text{nonCO2GHG}}$):

$$A_{\text{nonCO2GHG, nat}} = E_{\text{GWP100, nonCO2, nat}} - \frac{\Delta T_{\text{nonCO2, future, nat}} \cdot k_{\text{norm}}}{\text{TCRE}}$$

Where $\text{TCRE} = 0.00044959\ \text{K/Gt CO}_2$ (IPCC AR6 median) and $k_{\text{norm}} = 1.04$ (mean across 841 FaIR v2.2.4 ensemble configurations).

Transient Temperature Warming Response (ΔT):

$$\Delta T_{\text{scenario}} = E_{\text{cum, 2020–2050}} \cdot \text{TCRE} \cdot 1000 \quad [\text{mK}]$$

3.2 The 5-Dimensional Equity Matrix

The Hefaistos evaluates national allocations across five distinct equity paradigms:

Ability-to-Pay (PPP GDP):

$$B_{\text{ATP}} = f_{\text{GDP, PPP}} \cdot B_{\text{gl}} + A_{\text{LULUCF}} + A_{\text{nonCO2GHG}}$$

Equal Per-Capita:

$$B_{\text{Pop}} = f_{\text{pop}} \cdot B_{\text{gl}} + A_{\text{LULUCF}} + A_{\text{nonCO2GHG}}$$

Historical Responsibility (1990+):

$$B_{\text{Hist}} = f_{\text{pop}} \cdot B_{\text{gl, 1990–2050}} - E_{\text{cum, 1990–2019}} + A_{\text{LULUCF}} + A_{\text{nonCO2GHG}}$$

Climate / Heating Degree Days (HDD-Adjusted):

$$B_{\text{HDD}} = \left(f_{\text{pop}} \cdot \text{HDD_Factor}_{\text{nat}}\right) \cdot B_{\text{gl}} + A_{\text{LULUCF}} + A_{\text{nonCO2GHG}}$$

Trade / Consumption-Based Accounting:

$$B_{\text{Trade}} = B_{\text{nat}} \cdot \left(\frac{\text{Territorial Emissions}}{\text{Consumption Footprint}}\right)$$

4. EMPIRICAL VALIDATION & COMPARATIVE CASE STUDIES

The platform has been validated against published academic benchmarks (Partanen et al., 2026, Table 1) with **100% numerical precision**. Below is an empirical comparison of three fundamentally distinct economies analyzed by The Hefaistos:

Metric / Analytical Dimension	Finland (FIN) FI	Greece (GRC) GR	Brazil (BRA) BR
Clean Power Generation Share	96.2 % (Bio/Nuclear/Wind)	48.0 % (Wind/Solar)	90.4 % (Hydro/Wind/Solar)
Fossil Fuel Share in Power	3.8 %	52.0 % (Gas/Lignite)	9.6 %
Coal Phase-Out Projected Year	2026 (Rapid structural exit)	2026 (Lignite phase-out)	None (Marginal coal < 2%)
Overall Electrification Rate	28.2 %	25.7 %	20.4 %
IEA Baseline Emissions	40.2 Mt CO_2e	56.6 Mt CO_2e	408.2 Mt CO_2e (Energy only)
Non-CO_2 Adjustment (A_{nonCO2})	-0.436 Gt	-0.580 Gt	-29.675 Gt (High livestock methane)
LULUCF Adjustment (A_{LULUCF})	-0.438 Gt	0.000 Gt	0.000 Gt
Transient Warming Impact (ΔT)	+0.41 mK	+0.57 mK	+3.90 mK
Remaining Budget: Ability-to-Pay (PPP)	-0.654 Gt (Carbon Debit)	+1.667 Gt (Carbon Surplus)	-7.211 Gt (High non-CO_2 burden)
Remaining Budget: Equal Per-Capita	-0.218 Gt (Carbon Debit)	+0.637 Gt (Carbon Surplus)	-4.590 Gt (Debit)
Remaining Budget: Historical (1990+)	-0.351 Gt (Carbon Debit)	+0.240 Gt (Carbon Surplus)	**+9.227 Gt (SURPLUS CREDIT)**
Remaining Budget: HDD-Adjusted	**+0.077 Gt (SURPLUS CREDIT)**	+0.576 Gt (Carbon Surplus)	-7.099 Gt (Debit)
Remaining Budget: Trade-Adjusted	-0.161 Gt (Carbon Debit)	+0.522 Gt (Carbon Surplus)	-4.989 Gt (Debit)

Key Policy Insights Revealed:

Finland: Demonstrates that high clean power shares (96%) are insufficient on their own; northern heating demands (HDD) and managed forest sink accounting dominate national carbon space. When basic-needs heating is recognized, Finland's budget turns positive ($+0.077$ Gt).

Greece: Highlights the risk of fuel switching from lignite to fossil gas (37.5%), which mitigates CO_2 but creates non-CO_2 methane burdens (-0.58 Gt).

Brazil: Proves that emerging decarbonized power grids (90% clean) face entirely different policy frontiers: $70\%+$ **of emissions stem from agriculture, livestock methane, and land-use change**. Under historical responsibility, Brazil retains a substantial $+9.23$ Gt **carbon credit surplus**.

5. ACADEMIC OUTPUT PIPELINE & REPRODUCIBILITY

Every analysis execution in The Hefaistos produces a standardized, publication-grade **15-artifact research suite**:

Executive Summary (`executive_summary.md` / `.html`): High-level ministerial decision-maker briefing.

Policy Brief (`policy_brief.md` / `.html`): Actionable 0–2 yr, 2–5 yr, and 5+ yr reform schedules.

Peer-Review Summary (`peer_review_summary.md`): Methodological checks, sensitivity intervals, and data limitations.

LaTeX Methods (`methods_section.tex`): Journal-ready mathematical formulation with CES equations and FaIR parameters.

Quality Assurance Audit (`verification_report.json`): Strict verification gates ensuring zero numeric hallucinations.

Data Manifests (`compute_bundle.json`, `carbon_budget_summary.csv`): Fully reproducible machine-readable matrices.

6. TECHNICAL SPECIFICATIONS & ARCHITECTURE

Computational Backend: Python 3.12, FastAPI, Pydantic v2, NumPy, Pandas, xArray, SciPy, Pyomo.

Deterministic Execution: Strict separation between numerical compute engines (zero LLM in calculations) and natural language synthesis.

Storage & Episodic Memory: SQLite, ChromaDB episodic vector memory, and JSON/CSV data layers.

Frontend Presentation: Responsive, zero-dependency modern HTML5/CSS3 interface with live polling and interactive presentation views.

7. CONCLUSION & STRATEGIC IMPACT

The Hefaistos transforms climate policy analysis from siloed guesswork into an **integrated, physical, and equitable science**. By linking macroeconomic policy instruments directly to geophysical atmospheric outcomes and multi-criteria fairness rules, the platform empowers governments, international institutions, and research organizations to design climate pathways that are **economically viable, physically sound, and globally just**.

8. REFERENCES & METHODOLOGICAL FOUNDATIONS

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